12ELMOG3

SomeonecoolLovesMaths

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Problem

ABC is a triangle with incenter I. The foot of the perpendicular from I to BC is D, and the foot of the perpendicular from I to AD is P. Prove that $\angle BPD = \angle DPC$.

Solution. Let the incircle touch AC at E and AB at F. Let EF intersect BC at X.
We know $(X, D; B, C) = -1$.
Note that X lies on the polar of A so A must lie on the polar of X. Since D lies on the polar of X so AD must be the polar of X.
So, $XI \perp AD$ which implies that I, P, X are collinear and $\angle XPD = 90^\circ$. Since $(X, D; B, C) = -1$, we can conclude that PD bisects $\angle BPC$.